Assignment #1

Data Self Portrait: Refinement

Campbell Smith

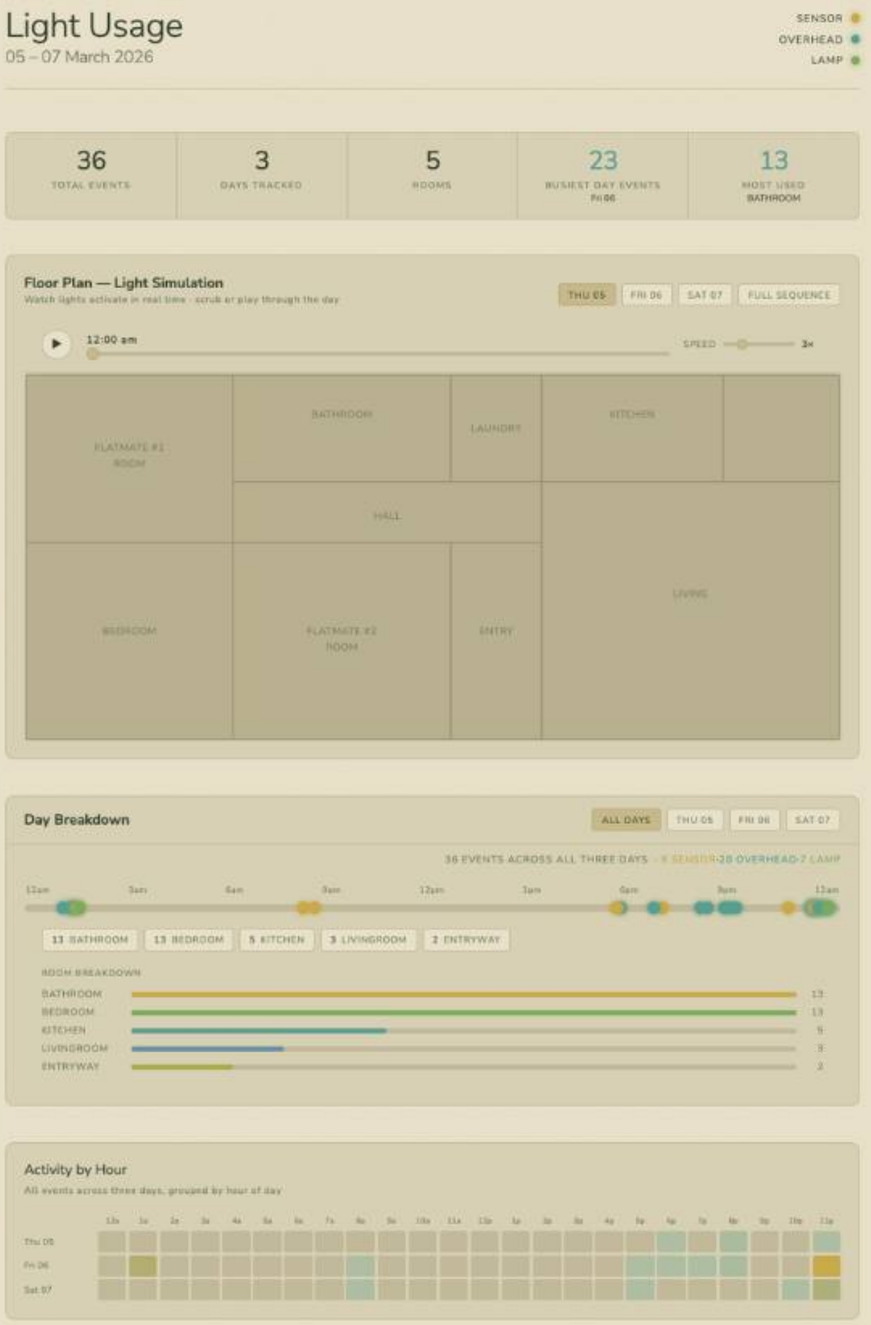
The Concept

The intention of this data collection was to see how light usaged revealed movement, and activity on a daily basis within my flat.

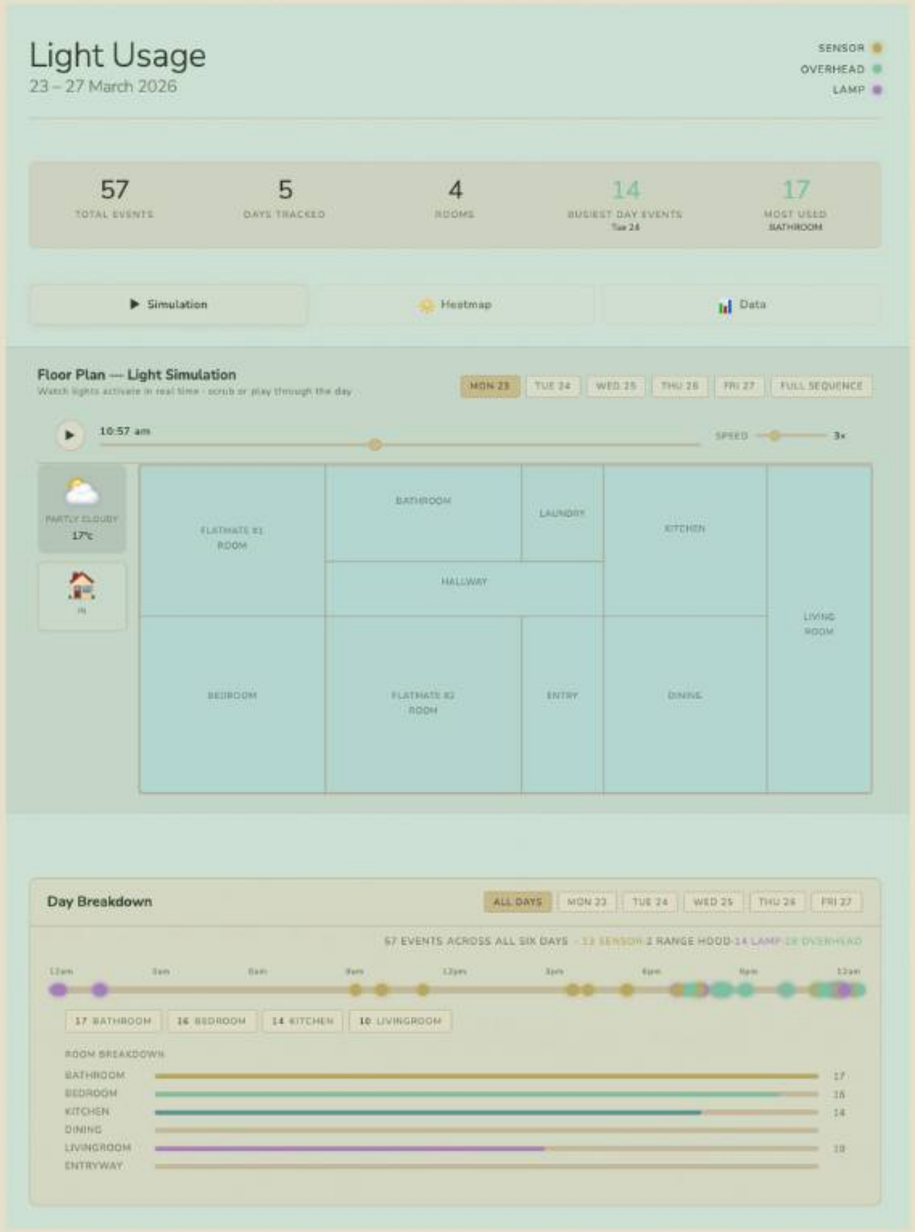
During the intial sketches I found out I was able to use Claude to create a simulation as my data was time and date based, and revolved my dashboard around this. In this extension of that dashboard I wanted to add in more insights about movement, external factors such as natural light and whether or not I was home, and a quick capture static visual to show the light usage.

As part of my final output I used Claude to create a app for the tracking of light usage via NFC tags, and iterated on the original dashboard.

https://claude.ai/public/artifacts/9eb9b16c-3523-422d-93e1-498edcca4568



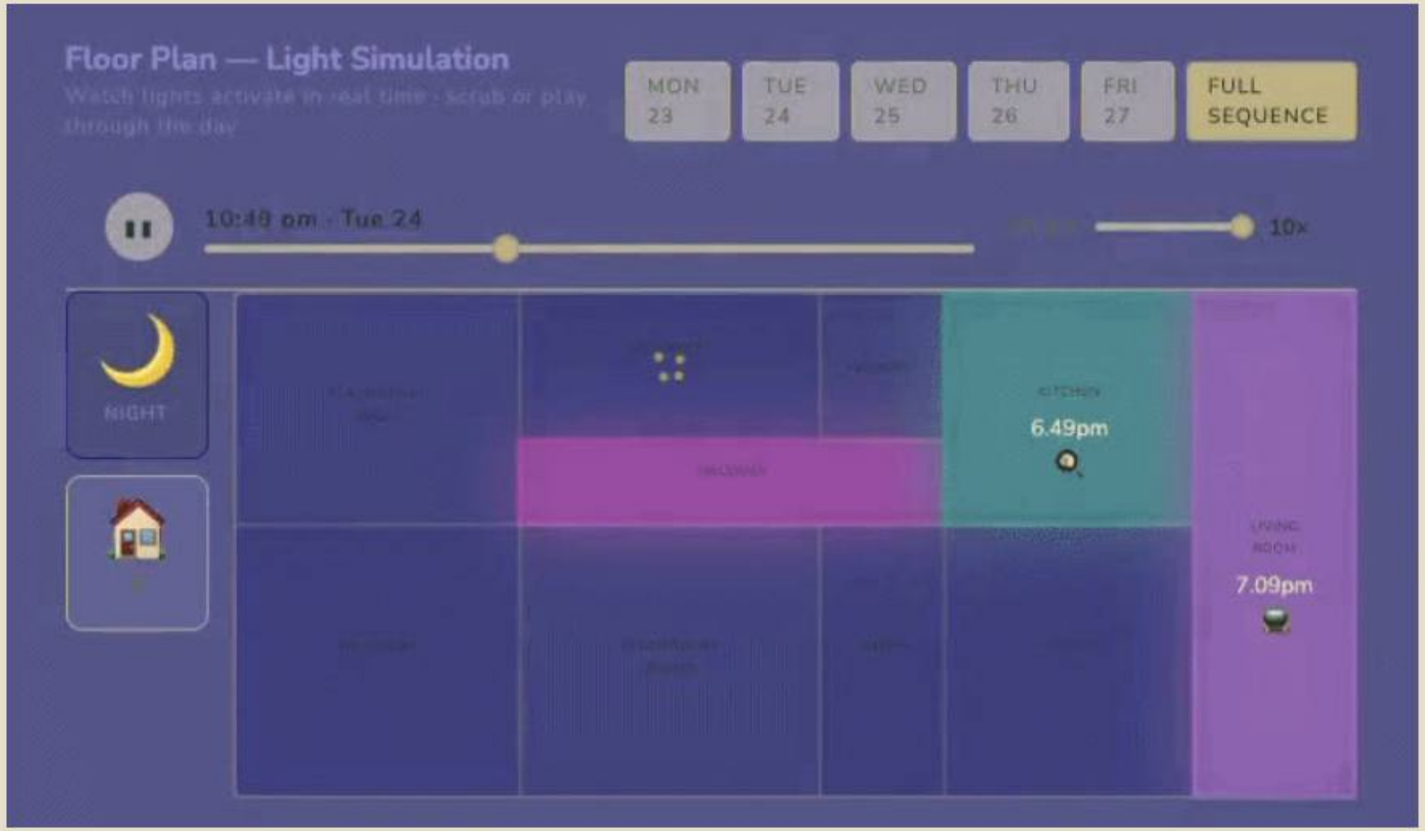
Final - Link to Artifact

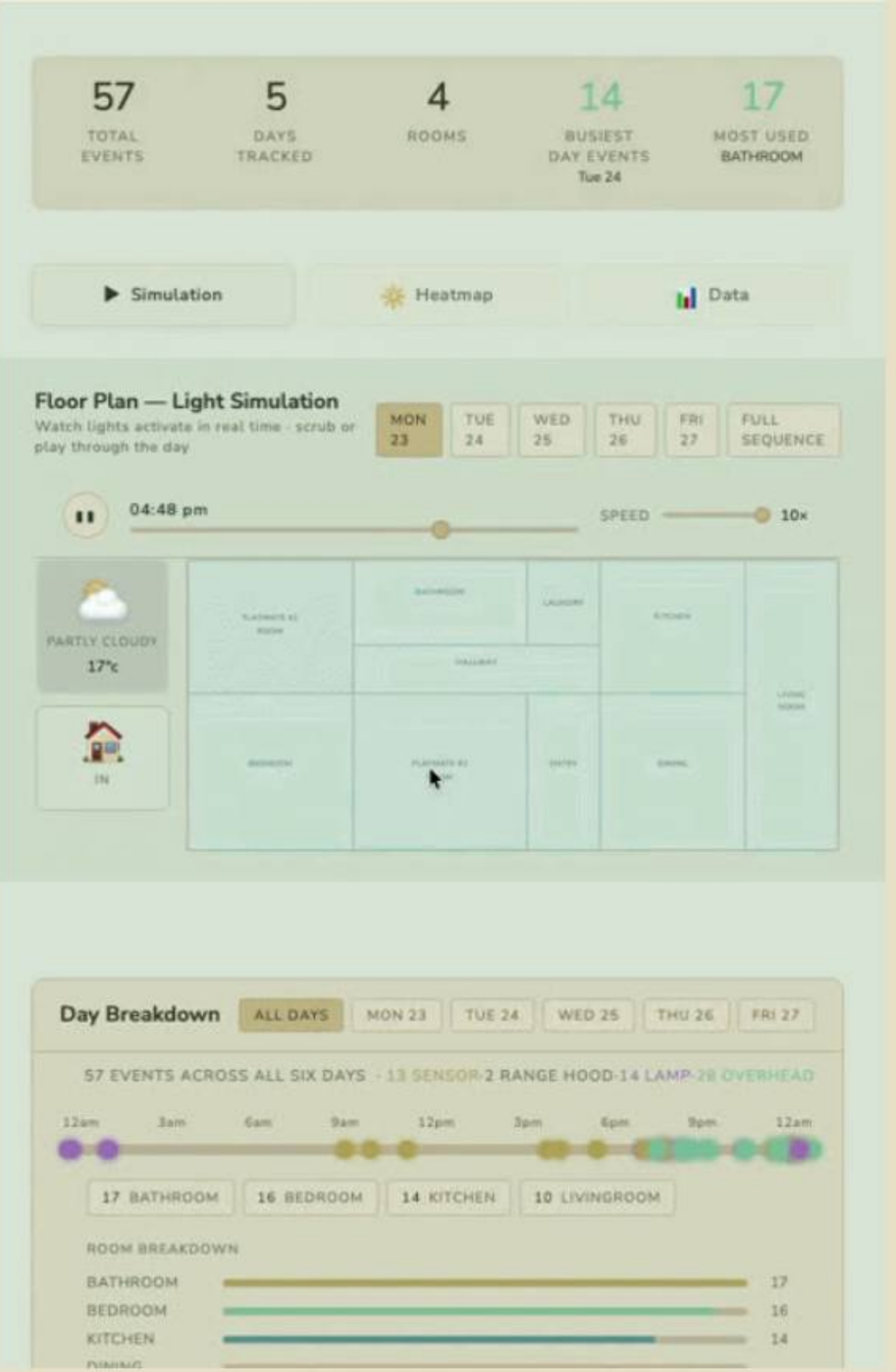






Final In Action - Link to Artifact





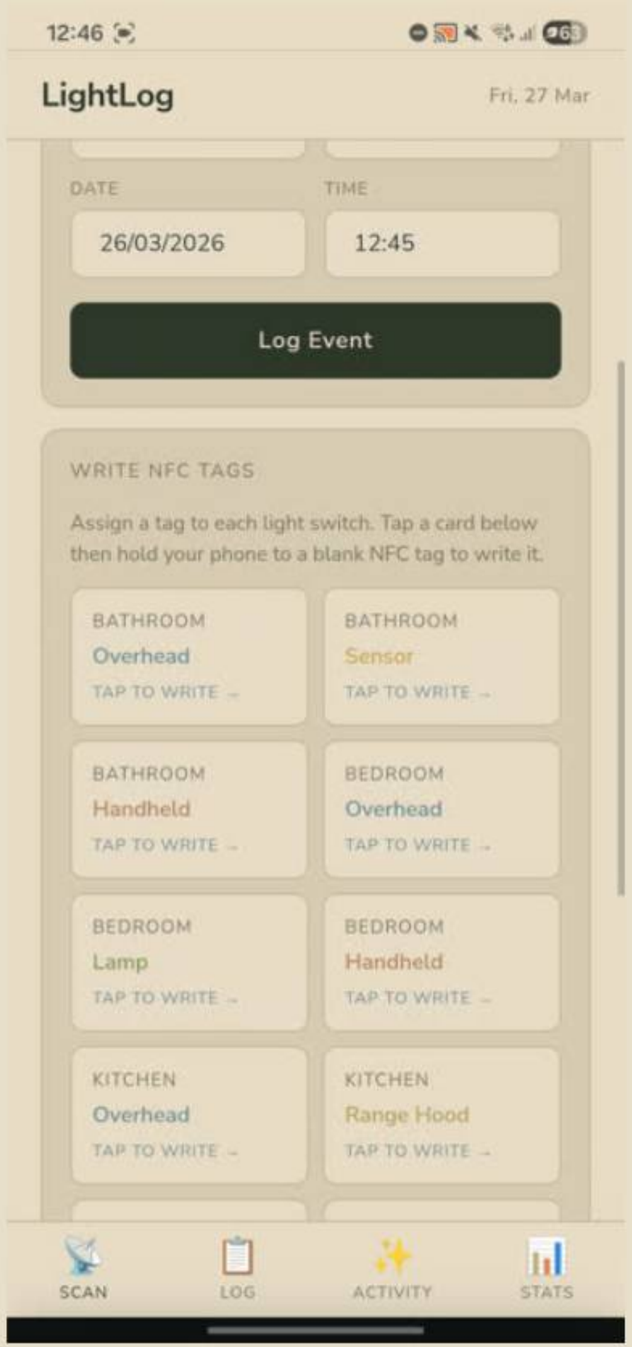
Process

To collect the data for this dashboard, I used Claude to create an app for my phone that set up an easy way to track my data. Using this app I integrated NFC tag placed around my flat where light switches were. Everytime I turned on a light, or entered a room where one of my flatmates had already turned on a light, I scanned the NFC tag and it was logged into the app. From the app under the stat's tab I was able to export my data straight into a csv file.

The App contains four tabs,
The scan tab where NFC tags can be written and scanned to input data, OR events can be manually logged.
The Log tab, where events can be checked, including edited if there was an accidental scan.
The Activity Tab, where the assumed events are logged, and can be edited.
And the stats tab, which contains a small screenshot of the data collected and a CSV can be downloaded of the logged data for direct uploading to the dashboard.



Link to Tracking App



Process

Once the data is imported from the app to the dashboard I was able to consume the data in a more visually appealing and interactive way.
During the sketching process I had already made the simulation, and the design was flat across the one page, as part of my development I added in activity to my tracking, estimated electricity cost/use, imported weather data, and also imported sunrise/sunset data for my flats area. One further addition to my data was adding in times in which I was out of the flat, I got this data posthumously from google which has location data on my phone.

Once these factors were added I split my dashboard into three areas, the simulation page, the heatmap, and the data. The simulation contains my original simulation with a few additional features I mentioned above. The heatmap section is the new interactive page it allows users to click on the room they are interested in, and a synopsis of the data pops up below, by pressing a button you can view all of the data in a glimpse.

Claude Use

App Dev

Intially I wasn't sure how I wanted to track the data, I knew I wanted to use NFC tags, as it would create less friction for taking note of every light turned on, but I wasn't sure how I wanted it to look. I fed this information into Claude and from there I did very minimal work on the app, only asking it to fix a few issues with code, and with the NFC tag writing. Once I had the app functional I hosted it on a Netfly website which made sure the data was saved online rather than locally on my phone.

Once I realised I wanted to try track activity I fed this idea into claude, and adjusted to activities I realistically use rooms for, altering from it's predictions, such as my bedroom lamp being used for sleep, not study which Claude assumed.

Visualisation

Claude was used mostly as a development tool, I didn't consider until I had completed the project that I could have used Claude more as a sounding board to ideate the final output.
A key discovery over the process of developing with Claude was it's limitations with animation. As part of the simulation section I tried to add in an element of movement, based on the last light turned on a cluster of dots moves from room to room, assuming the movement taken around the house, intially the animation had the dots jumping around at a fast pace and I had to correct it to follow a determined path to become less jarring. Even with these changes the dots do appear in rooms where the light isn't on as Claude predicts the movement based on last interaction, I ended up keeping this choice as movement was another element that was hard to track with Claude.

Reflection

The final dashboard reveals the movement within my flat as a easy to consume simulation that shows off how movement, activity, and light are connected in home life.

I think the most influencial of the readings on this project was the dear data project, the capturing of data in one smaller easier to consume visual was the goal of using the simulation element, and I believe I achieved that, the main straif from the work shown in the collection is the inherint interactivity and animation of the light log simulation, that comes from the work being digital rather than a physical graphic. The other influencial element from the reading in the final output would have been the discussion in the Seductions of Clarity on gamification, the awareness of scanning each light towards the end of my data tracking started to become evident, as I would run around the house scanning each NFC tag before going to sleep, and the data collection itself became a task knowing the end visual would be more appealing from tracking evvery light switch even if I had only turned a light on for a minute.

From the work produced I am proud of the elements I was able to create in Claude, I'd like to have added in more animated elements, but am aware from the movement element I did add that Claude was limited in this aspect.

Further developments to me for this project would be incorporating smart LED bulbs, the app for tracking data, and the dashboard itself. Ideally the dashboard and simulation would be a feature on the app, and could be a reflection of light activity activly and posthumously.

Incorperating smart LEDs they could be controlled via the app, or controlled manually, and the data would port over automatically. The simulation could also be viewed live, this way I'd be able to note if there was a light on that wasnt nessasary.